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Spring 26

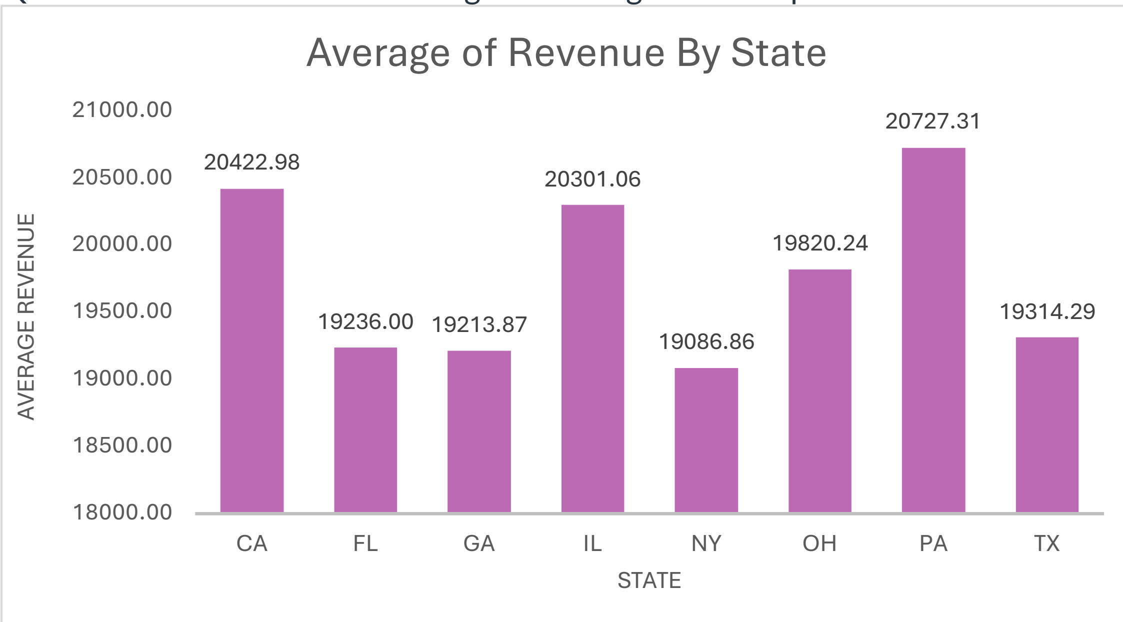
BAN 650

Assignment 1

Assignment Open Date: January 28, 2026

Assignment Due Date: February 11, 2026

Task 1:
Question: Which State has the highest average revenue per order ?



Why this chart type was selected
A column chart (vertical bar chart) was chosen because:

- It enables easy comparison of average revenues across discrete categories (states).
- Bar length offers a clear visual representation of differences, making it intuitive for managers to spot variations quickly.
- This format is ideal when showing categorical vs. numerical data — here, each state is a category with a unique revenue metric.

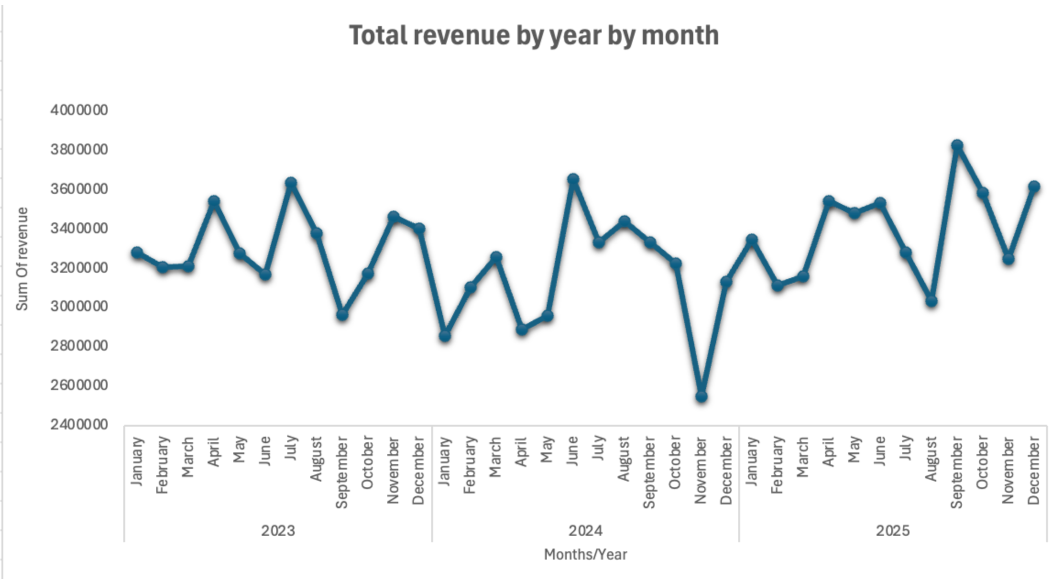
Insight the visualization communicates
This visualization highlights how average revenue varies by state:

- Pennsylvania (PA) has the highest average revenue (~20,727), followed by California (CA) and Illinois (IL).
- New York (NY) records the lowest average (~19,086), suggesting potential performance issues or lower market potential there.
- The moderate range (about 19,000–21,000) indicates relatively small variation, so managers can focus more on factors driving the top-performing states than on addressing severe underperformance.

A manager could interpret this to:
- Investigate what drives strong revenue in PA, CA, and IL (e.g., customer demographics, sales strategies, or product demand).
- Prioritize resource allocation or marketing efforts in lower-revenue states.

How best practices were applied for clarity
- Clear chart title: "Average of Revenue by State" immediately states the purpose of the visualization.
- Proper axis labels: “State” (x-axis) and “Average Revenue” (y-axis) make it easy to understand what each axis represents.
- Consistent scale: The y-axis uses a uniform numeric scale starting near 18,000 and ending at 21,000, maintaining proportional visual comparison.
- No unnecessary decoration: Minimalistic design — no 3D effects or grid clutter — ensures the data stands out.
- Data labels: The exact values above bars help managers reference specific numbers without estimating from the axis.

Task 2:
Question: What is the monthly revenue year on year (2023-2025)



Why this chart type was selected

A line chart was chosen because:

- It connects sequential monthly data points across 36 months, revealing progression patterns that column charts would fragment into disconnected snapshots.
- The continuous line makes fluctuation cycles immediately visible, highs and lows emerge naturally versus static monthly comparisons.
- Perfect for time-series data where managers need to understand month-to-month rhythm over categorical breakdowns.

Insight the visualization communicates

This visualization reveals revenue volatility across 2023-2025:

- Annual totals fluctuate 2023: $39.7M → 2024: $37.8M (-5%) → 2025: $40.8M (+8%) with no consistent directional trend.
- Monthly revenue swings 2.5M-3.8M around 3.2M average without predictable seasonal patterns.
- Consistent pattern of revenue drops after monthly peaks across all years.

A manager could interpret this to:

- Investigate why there is always a drop in revenue after a peak.
- Monitor monthly volatility drivers (sales reps, products, regions) rather than seasonal cycles.
- Focus improvement efforts on stabilizing post-peak declines.

How best practices were applied for clarity

- Clear chart title: "Total Revenue by Year/Month" immediately communicates time-series analysis purpose.
- Proper axis labels: "Month/Year" (x-axis) clearly defines temporal progression; Revenue scale (y-axis) uses currency formatting.
- Consistent scale: Y-axis spans actual data range (~2M-4M) focusing attention on realistic variation patterns.
- No unnecessary decoration: Single clean line with minimal gridlines eliminates chart junk, data tells the story.
- Strategic labeling: Month/year combinations provide precise time reference without overcrowding the x-axis.